A Project report on

Neural Style Transfer

A Dissertation submitted to JNTU Hyderabad in partial fulfillment of the academic requirements for the award of the degree.

Bachelor of Technology

in

Computer Science and Engineering

Submitted by

M.PAVAN
(20H51A05H9)

P.NEETHIKA
(20H51A05J2)

Under the esteemed guidance of

Dr. Siva Skandha Sanagala
(Associate Professor and HOD)

Department of Computer Science and Engineering

CMR COLLEGE OF ENGINEERING & TECHNOLOGY

(UGC Autonomous)

*Approved by AICTE *Affiliated to JNTUH *NAAC Accredited with A^+ Grade

KANDLAKOYA, MEDCHAL ROAD, HYDERABAD - 501401.

2020- 2024

CMR COLLEGE OF ENGINEERING & TECHNOLOGY

KANDLAKOYA, MEDCHAL ROAD, HYDERABAD – 501401

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the Major Project Phase I report entitled **"NEURAL STYLE TRANFER "** being submitted by M.Pavan (20H51A05H9), P.Neethika (20H51A05J2) in partial fulfillment for the award of **Bachelor of Technology in Computer Science and Engineering** is a record of bonafide work carried out his/her under my guidance and supervision.

The results embody in this project report have not been submitted to any other University or Institute for the award of any Degree.

Dr. Siva Skandha Sanagala
Associate Professor and HOD
Dept. of CSE

ACKNOWLEDGEMENT

With great pleasure we want to take this opportunity to express my heartfelt gratitude to all the people who helped in making this project work a grand success.

We are grateful to **Dr. Siva Skandha Sanagala,** Associate Professor and Head of Department, Department of Computer Science and Engineering for his valuable technical suggestions and guidance during the execution of this project work ans is the major driving forces to complete my project work successfully.

We are very grateful to **Dr. Devadasu** , Dean-Academics, CMR College of Engineering and Technology, for his constant support and motivation in carrying out the project work successfully.

We are highly indebted to **Major Dr. V A Narayana,** Principal, CMR College of Engineering and Technology, for giving permission to carry out this project in a successful and fruitful way.

We would like to thank the **Teaching & Non- teaching** staff of Department of Computer Science and Engineering for their co-operation

We express our sincere thanks to **Shri. Ch. Gopal Reddy**, Secretary, CMR Group of Institutions, for his continuous care.

Finally, we extend thanks to our parents who stood behind us at different stages of this Project. We sincerely acknowledge and thank all those who gave support directly and indirectly in completion of this project work.

M.Pavan 20H51A05H9
P.Neethika 20H51A05J2

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	LIST OF FIGURES	ii
	LIST OF TABLES	iii
	ABSTRACT	iv
1	**INTRODUCTION**	**1**
	1.1 Problem Statement	2
	1.2 Research Objective	2
	1.3 Project Scope and Limitations	3
2	**BACKGROUND WORK**	**5**
	2.1. A Neural Algorithm of Artistic Style	6
	2.1.1. Introduction	6
	2.1.2. Merits, Demerits and Challenges	6
	2.1.3. Implementation	8
	2.2. Unpaired Image-Image Translation	9
	2.2.1. Introduction	9
	2.2.2. Merits, Demerits and Challenges	9
	2.2.3. Implementation	11
	2.3. Perceptual Losses for Real-Time Style Transfer	11
	2.3.1. Introduction	12
	2.3.2. Merits, Demerits and Challenges	12
	2.3.3. Implementation	13
3	**RESULTS AND DISCUSSION**	**16**
	3.1. Performance metrics	17
4	**CONCLUSION**	20
	4.1 Conclusion	21
	REFERENCES	**22**
	GitHub Link	**24**

List of Figures

FIGURE NO.	TITLE	PAGE NO.
3.1	Input Image	19
3.2	Style Image	19
3.3	Generated Image	19

List of Tables

FIGURE NO.	TITLE	PAGE NO.
2.1	Background Work .	16

ABSTRACT

The problem statement for the proposed project, Neural Style Transfer, is to design and implement an efficient deep learning model for style transfer, aiming to combine the artistic style of one image with the content of another image (content imag). We came with a solution to design and implement an efficient deep learning model for style transfer. This model takes a content image ‘C’ and a style image ‘S’ and merge both images to produce a new image that contains the content of image ‘C’ and the style of image ‘S’. This can be implemented by neural style transferring using deep learning which involves transferring the artistic style of one image onto the content of another, resulting in creative and visually appealing images as outputs. It will be implemented through Convolutional Neural Networks (CNNs), utilizing pre-trained models. The process involves extracting features from both content and style images through pre-trained network, pre-processing of images, optimizing a generated image to minimize content and style loss and iteratively refining the results.

CHAPTER 1
INTRODUCTION

CHAPTER 1
INTRODUCTION

1.1. Problem Statement

In the dominance of digital media and creative content generation, there is a growing need for automated tools that can seamlessly blend the artistic style of one image with the content of another. The problem at hand is to develop a robust and efficient style transfer algorithm capable of preserving the content of an input image while applying the desired artistic style from a reference image. This project aims to address the challenges of creating a user-friendly, real-time style transfer solution that produces high-quality results across various domains, including art, photography, and graphic design.

1.2. Research Objective

The research objective for neural style transfer can be framed as follows:

- To develop neural style transfer, the primary objective is to develop and investigate techniques, algorithms, and models that enhance the efficiency, flexibility, and artistic quality of neural style transfer methods. This includes:
- Improved Real-time Performance: Develop faster and more efficient neural style transfer models that can be applied in real-time or near-real-time, enabling applications in video processing, live streaming, and interactive artistic tools.
- Content and Style Control: Research methods that allow for more precise control over the artistic style and content of the generated images, enabling users to customize the output to a greater extent.
- Multi-Modal Style Transfer: Explore the extension of neural style transfer to different modalities, such as audio, text, or 3D models, to enable cross-domain artistic synthesis and new creative applications.
- Evaluation and Metrics: Develop standardized evaluation metrics for assessing the quality and effectiveness of neural style transfer methods, both qualitatively and quantitatively.

- Ethical and Fair Use: Consider the ethical implications of neural style transfer, including issues related to copyright, privacy, and responsible use, and propose guidelines and best practices.

1.3 Project Scope and Limitations

The project scope and limitations of a neural style transfer project are essential to define to ensure that the research or development effort is focused. Here are some considerations for the project scope and limitations.

Project Scope:

Artistic Style Transfer: The primary scope of the project is to implement and optimize neural style transfer techniques for transferring artistic styles from one image to another. This can include various artistic styles or other visual styles. Real-Time and Batch Processing: Decide whether the project will focus on real-time applications (e.g., live video processing) or batch processing (e.g., applying style transfer to a collection of images). Define the scope accordingly. Evaluation Metrics: Define the metrics and evaluation criteria that will be used to assess the quality and effectiveness of the style transfer results.

Project Limitations:

- Artistic Style Limitations: Acknowledge that not all artistic styles may be compatible with neural style transfer techniques. Some styles may not transfer effectively or may require specialized approaches.
- Artistic Quality: Neural style transfer has inherent limitations in achieving the exact quality and intricacies of human-created art. The project should recognize that the results are artistic interpretations rather than perfect replicas.
- Content Constraints: The content of the content and style images should be compatible for effective style transfer. The project should be aware of limitations when applying styles to specific content types.
- Overfitting: Neural style transfer models can be prone to overfitting the style, which may result in artifacts. Implement regularization techniques to mitigate this, but acknowledge that some level of overfitting may still occur.

- Ethical Considerations: Be aware of ethical concerns, such as copyright and privacy issues, when using real-world data for style transfer. Respect these limitations and ensure responsible usage.
- Generalization: Neural style transfer models may not generalize well to all types of content or styles. The project should acknowledge potential limitations in the generalizability of the model.
- User Expertise: If the project involves user customization, recognize that users may have varying levels of expertise in using the tool. The system should be user-friendly and not overly complex.

Cultural and Contextual Factors: Acknowledge that the interpretation of artistic styles can vary across cultures and contexts, and the project may not capture all nuances. Defining the project scope and limitations is crucial for setting realistic expectations and ensuring the project's success. It helps in managing resources, time, and the quality of the results while addressing potential challenges and constraints.

CHAPTER 2
BACKGROUND WORK

CHAPTER 2
BACKGROUND WORK

2.1Neural Algorithm of Artistic Style

2.1.1 Introduction

In fine art, especially painting, humans have mastered the skill to create unique visual experiences through composing a complex interplay between the content and style of an image. Thus far the algorithmic basis of this process is unknown and there exists no artificial system with similar capabilities. However, in other key areas of visual perception such as object and face recognition near-human performance were recently demonstrated by a class of biologically inspired vision models called Deep Neural Networks. Here we introduce an artificial system based on a Deep Neural Network that creates artistic images of high perceptual quality. The system uses neural representations to separate and recombine content and style of arbitrary images, providing a neural algorithm for the creation of artistic images. Moreover, in light of the striking similarities between performance-optimized artificial neural networks and biological vision, our work offers a path forward to an algorithmic understanding of how humans create and perceive artistic imagery.

2.1.2 Merits, Demerits and Challenges

The neural algorithm of artistic style, as proposed by Leon A. Gatys, Alexander S. Ecker, and Matthias Bethge in their 2015 paper titled "A Neural Algorithm of Artistic Style" (published in IEEE), has certain merits and demerits.

Merits:

- High-Quality Artistic Transformations: This approach is capable of producing high-quality artistic transformations that closely resemble the style of famous artworks. It allows for the generation of visually appealing and creative images.
- Content Preservation: The method effectively preserves the content of the input image while applying the style of a reference image, ensuring that the final output remains recognizable and meaningful.

- Flexibility: It is highly versatile and can be applied to a wide range of content and styles, making it a valuable tool for artists and designers. You can apply various artistic styles to any image.
- End-to-End Deep Learning: Leveraging deep learning, this approach can learn and adapt to different styles and content automatically, without the need for manual feature engineering.

Demerits:

- Computational Complexity: The neural algorithm of artistic style can be computationally expensive and may require powerful hardware, such as GPUs, for practical usage. This limits its real-time or large-scale applicability.
- Training: The method involves training a convolutional neural network (CNN) on a particular style image, which can be time-consuming and may not work well for less common or less well-defined artistic styles.
- Parameter Tuning: Achieving the desired results often involves fine-tuning hyperparameters, which can be a time-consuming process. Adjusting these parameters for optimal output can be challenging for non-experts.
- Limited Realism: While the method produces visually appealing images, it doesn't inherently enhance the realism of the content. The output can sometimes look more like a painting or artwork than a realistic photograph.
- Legal and Ethical Concerns: Applying the style of famous artworks to images can raise legal and ethical concerns, particularly related to copyright and intellectual property rights.

In summary, the neural algorithm of artistic style proposed by Gatys, Ecker, and Bethge is a powerful technique for creating visually appealing images by combining content and style. It shares similar merits and demerits with the original approach, including computational challenges, the need for parameter tuning, and the potential for loss of realism and semantic information. Despite these limitations, it has had a significant impact on the field of computer vision and art generation.

2.1.3. Implementation of A Neural Algorithm of Artistic Style

Implementing the Neural Algorithm of Artistic Style, as proposed by Leon A. Gatys, Alexander S. Ecker, and Matthias Bethge, typically involves using deep learning frameworks like TensorFlow or PyTorch. Below, I'll provide a high-level outline of the steps involved in implementing this algorithm. Please note that this is a simplified version for educational purposes, and a complete implementation may require additional details and optimizations.

- The initial step is preprocessing where; the content image and style image are loaded. Resize both images to the same dimensions. The pixel values of both images are normalized. Images are converted into tensors that can be processed by the deep learning framework.
- A pre-trained model is choosing convolutional neural network. You can load these models from TensorFlow Hub or torch vision models in PyTorch.
- Define Content and Style Layers: Identify the layers in the chosen model that will be used to extract content and style information. Typically, earlier layers are used for style, and later layers are used for content.
- Loss Functions: Define the content loss, which is the Mean Squared Error (MSE) between the feature maps of the content image and the generated image at the selected content layers.
- Define the style loss, which is computed by comparing the Gram matrices of the feature maps at the selected style layers.
- Generate the Target Image: Initialize a target image (usually as a copy of the content image).
- Create a TensorFlow Variable or PyTorch Tensor to represent the target image, which allows you to optimize its pixel values.
- Optimization: Use an optimization algorithm to minimize a combination of content loss and style loss. You can also add a total variation loss to reduce noise in the generated image.
- Perform several iterations of the optimization process, updating the target image to minimize the combined loss.

- Postprocessing: Once you've obtained the optimized target image, deformalize it by reversing the normalization process.

REMARKS: This paper revolutionized the field of neural style transfer by introducing a method to combine content and style in images using deep learning [2]

2.2 Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks

2.2.1 Introduction

Image-to-image translation is a class of vision and graphics problems where the goal is to learn the mapping between an input image and an output image using a training set of aligned image pairs. However, for many tasks, paired training data will not be available. We present an approach for learning to translate an image from a source domain X to a target domain Y in the absence of paired examples. Our goal is to learn a mapping $G : X \rightarrow Y$ such that the distribution of images from G(X) is indistinguishable from the distribution Y using an adversarial loss. Because this mapping is highly under-constrained, we couple it with an inverse mapping $F : Y \rightarrow X$ and introduce a cycle consistency loss to enforce $F(G(X)) \approx X$ (and vice versa). Qualitative results are presented on several tasks where paired training data does not exist, including collection style transfer, object transfiguration, season transfer, photo enhancement, etc. Quantitative comparisons against several prior methods demonstrate the superiority of our approach.

2.2.2 Merits, Demerits and Challenges

Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks by Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A. Efros, introduces a method for translating images from one domain to another without the need for paired training data. The primary technique used in this paper is CycleGAN, which leverages cycle-consistency to enforce a bijective mapping between two domains. Here are the merits, demerits, and challenges associated with unpaired image-to-image translation using Cycle-Consistent Adversarial Networks (CycleGAN):

Merits:

- No Paired Data Required: The most significant advantage of CycleGAN is that it can perform image-to-image translation without the need for paired data (i.e., images from the source domain and their corresponding images in the target domain). This makes it a more versatile and practical solution since collecting paired data can be expensive and time-consuming.
- Applicability: CycleGAN is versatile and can be applied to various image translation tasks, including style transfer, season transfer, and more. It can also be adapted for other domains beyond images.
- Generative Adversarial Network (GAN) Framework: CycleGAN is based on GANs, which have shown their effectiveness in generating high-quality images. This ensures that the generated images have a high degree of realism.

Demerits:

- Limited Control: CycleGAN may not provide fine-grained control over the translation process. While it can learn to perform image translation, it may not allow users to specify precise attributes or features to be translated, limiting its use in certain applications.
- Quality and Consistency: The quality and consistency of the generated images can vary. Some images may look realistic, while others may not be as convincing, depending on the complexity of the translation task.
- Computational Resources: Training a Cycle GAN model can be computationally intensive and time-consuming, especially for high-resolution images and complex translation tasks. This may require powerful hardware and substantial training time.
- The challenges to this method include: Mode Collapse where GAN”s including those which are used in Cycle GAN can suffer from mode collapse where the generator produces a limited set of similar images. Another challenge is that Data Augmentation where processing data set and augmenting it to for better results can be tough. Assessing the quality of generated images is a also a greater challenge in image-to-image translation.

2.2.3 Implementation

Network Architecture : They have adopted an architecture for generative networks. which have shown impressive results for neural style transfer and superresolution. This network contains three convolutions, several residual blocks, two fractionally-strided convolutions with stride 1 2 , and one convolution that maps features to RGB. They used 6 blocks for 128 × 128 images and 9 blocks for 256×256 and higher-resolution training images. They have used an instance normalization. For the discriminator networks they used 70 × 70 PatchGANs, which aim to classify whether overlapping image patches are real or fake.

REMARKS: It addresses the challenging problem of translating images from one domain to another without requiring paired training data, which is significant Advancement in the field of image-to-image translation. [1]

2.3 Perceptual Losses for Real-Time Style Transfer and Super-Resolution

2.3.1 Introduction

They have considered image transformation problems, where an input image is transformed into an output image. Recent methods for such problems typically train feed-forward convolutional neural networks using a per-pixel loss between the output and ground-truth images. Parallel work has shown that high-quality images can be generated by defining and optimizing perceptual loss functions based on high-level features extracted from pretrained networks. They have combine the benefits of both approaches, and propose the use of perceptual loss functions for training feed-forward networks for image transformation tasks. It show results on image style transfer, where a feed-forward network is trained to solve the optimization problem proposed by Gatys et al in real-time. Compared to the optimization-based method, our network gives similar qualitative results but is three orders of magnitude faster. They also experiment with single-image super-resolution, where replacing a per-pixel loss with a perceptual loss gives visually pleasing results.

2.3.2 Merits, Demerits and Challenges

Merits:

- Real-Time Style Transfer: The method allows for real-time style transfer, which means that it can be applied to images or videos in a live or interactive setting. This is a significant advantage for applications like video games, augmented reality, and live video processing.
- Style Preservation: The approach is effective at preserving the artistic style of the reference image during style transfer. This is important for applications where the style is a key component, such as art generation and branding.
- Generalization: The technique can be applied to a wide range of style transfer and super-resolution tasks without needing specific models or training for each task. This makes it more versatile and adaptable to various applications.
- Low-Level Details: In the case of super-resolution, the approach can effectively recover fine details and textures in images, making it useful for enhancing image quality.

Demerits:

- Resource Intensive: Implementing perceptual loss methods can be computationally intensive, particularly during training. Pre-trained deep neural networks like VGG are required, and training may demand powerful hardware.
- Complex Implementation: Implementing perceptual loss-based style transfer and super-resolution can be more complex than traditional techniques. It requires a good understanding of deep learning frameworks and often involves custom code.
- Overfitting Risk: Like other deep learning methods, there's a risk of overfitting, especially if the dataset used for training is limited or not diverse. Careful regularization and data augmentation are required.
- Model Dependency: The technique relies on pre-trained deep neural networks for feature extraction. Changes or advancements in these networks may affect the performance and stability of the method.

- Lack of Control: Perceptual loss methods may not offer fine-grained control over the style transfer or super-resolution process. Users have limited direct control over specific attributes or elements of the output.
- Hyperparameter Tuning: Tuning hyperparameters in the loss function and the optimization process can be challenging and time-consuming.

2.3.3. Implementation

The implementation of "Perceptual Losses for Real-Time Style Transfer and Super-Resolution" proposed by Justin Johnson, Alexandre Alahi, and Li Fei-Fei involves several steps

- Neural Network Architecture: The first step in implementing perceptual losses for style transfer and super-resolution is to define the neural network architecture. In the original paper, a pre-trained VGG network is commonly used. VGG is well-suited for its ability to capture high-level features from images, which will be used for perceptual loss computation. Depending on the task and the specific architecture used, the layers from which features are extracted can vary.
- Feature Extraction:With the selected neural network in place, the next step involves feature extraction. This is done by feeding both the reference image (for style transfer) or the low-resolution image (for super-resolution) and the target image (generated) through the network. Intermediate feature maps are extracted from specific layers of the network to capture information about texture, style, and content.
- Perceptual Loss Calculation:The perceptual loss is then computed using the extracted features. This loss typically comprises three components: content loss, style loss, and total variation loss. The content loss measures the similarity between the feature maps of the generated image and the reference image. The style loss quantifies the difference in texture and style between the generated image and the reference image. The total variation loss helps smooth out the final output.

- Optimization: Once the perceptual loss is computed, it is used as the objective function for optimization. The goal is to minimize this loss by adjusting the parameters of the generated image. This is often achieved using gradient-based optimization techniques like stochastic gradient descent (SGD) or its variants. The process involves iteratively updating the generated image to make it more similar to the reference image in terms of content and style.
- Hyperparameter Tuning: Fine-tuning the hyperparameters is a crucial aspect of the implementation. These hyperparameters include the learning rate, the weighting of the content and style losses, the number of iterations, and regularization terms. Careful tuning is necessary to achieve the desired balance between content preservation and style transfer.
- Post-Processing: Depending on the application and the specific neural network architecture used, some post-processing steps may be needed. For example, in the case of super-resolution, additional techniques like bicubic interpolation may be used to upscale the generated image to the desired resolution.
- Deployment and Real-Time Processing: The final step involves deploying the trained model for real-time processing. This may require optimizing the implementation for speed and efficiency to achieve real-time or interactive performance, which could involve optimizations like parallelization or using hardware accelerators.[3]

TABLE:

Year of Publication	Author Name	Name of Journal	Title of Paper	Technology Used	Technique Used
2015	Leon A. Gatys, Alexander S. Ecker,and Matthias Bethge	arXiv	A Neural Algorithm of Artistic Style	CNNs, Feature Extraction, Preprocessing and Post-processing	Content and style separation, Loss functions, Optimization, Feature Maps
2017	Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A. Efros	paperswithcode	Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks	CycleGAN, CNNs	GANs, Cycle-Consistency Loss, Adversarial Loss

Dataset	Performance Evaluation	Key Contributions	Type of Technique	Your View on the Paper	Architecture(if any)
ImageNet	Artistic Effectiveness, Customizability	Use of CNNs, Artistic Applications, Creative Possibilities	Deep learning based image processing	This paper revolutionized the field of neural style transfer by introducing a method to combine content and style in images using deep learning.	
CycleGAN Horse-to-Zebra	Visually appealing images, Consistency, Quality	Cycle-Consistency for Unpaired Image Translation	CycleGAN	It addresses the challenging problem of translating images from one domain to another without requiring paired training data, which is significant Advancement in the field of image-to-image translation.	Novel Architectural Design

Fig 2.1 Background Work

CHAPTER 3
RESULTS AND DISCUSSION

CHAPTER 3

RESULTS AND DISCUSSION

3.1 Performance Metrics

Evaluating the performance of neural style transfer can be somewhat subjective since it primarily deals with the aesthetics of generated images. However, there are a few metrics and methods use to assess the quality of neural style transfer results:

- Content Loss: These measures how closely the content of the generated image matches the content image. It's often calculated as the mean squared error (MSE) or the feature-wise cosine distance between feature maps from a pre-trained convolutional neural network.
- Perceptual Loss: Similar to content loss, it considers feature differences in deeper layers of a network to capture higher-level content. It's often used in combination with style loss.
- Style Loss: Measures how well the style of the generated image matches the style image. It is computed by comparing the Gram matrices of feature maps from pre-trained networks.
- Total Variation (TV) Loss: It's a regularization term that encourages spatial smoothness in the generated image. It can help reduce artifacts in the output.
- Human Evaluation: Often, the best way to assess the quality of style transfer is through human judgment. Conduct user studies or surveys to get feedback from people on the generated images.
- Comparative Evaluation: Compare the style transfer results with alternative methods or baseline techniques to demonstrate the superiority of your approach.
- Execution Time: Measure the time it takes to perform the neural style transfer, as this can be an important factor in practical applications.
- Style Diversity: If your neural style transfer method allows for multiple styles to be applied, you can measure the diversity of styles in the generated images.

- Consistency in Video or Sequences: If you are applying style transfer to videos or image sequences, you can measure the temporal consistency to ensure smooth transitions between frames.
- Depending on the specific use case, you might need to design custom metrics that capture certain desired features, textures, or objects in the generated images.

It's important to note that neural style transfer is often more of an art than a science, and the choice of metrics can depend on the specific goals of the project. In many cases, a combination of different metrics and human judgment is used to evaluate the performance comprehensively. Additionally, practical applications and user preferences often play a significant role in determining the success of a neural style transfer method. In conclusion, neural style transfer has proven to be a powerful and creative application of deep learning. It has the potential to revolutionize art and design while finding applications in various other fields. However, it is not without its challenges, including computational complexity and ethical concerns. As research in this area continues, we can expect further advancements that will make neural style transfer more accessible and impactful in the creative and commercial realms

Here is an example of how neural style transfer works by combining the content of one image with the content of another image

Fig 3.1 Input Image

Fig. 3.2 Style

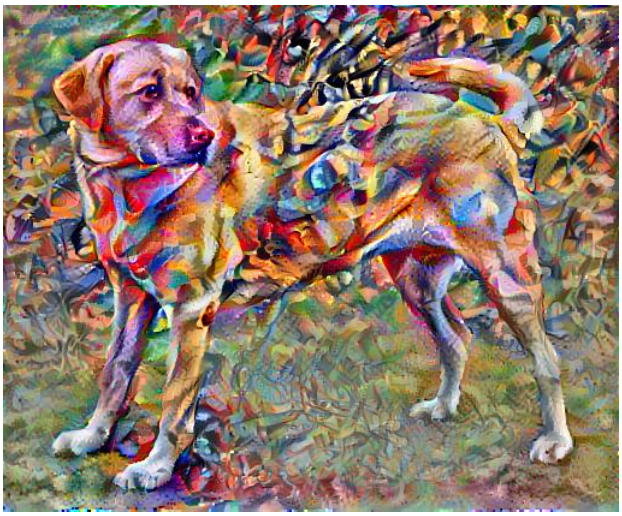

Fig 3.3 Generated Image

CHAPTER 4
CONCLUSION

CHAPTER 4
CONCLUSION

In conclusion, the project on neural style transfer has provided valuable insights into the field of computer vision and deep learning. Neural style transfer, a technique that combines the content of one image with the artistic style of another, has numerous practical applications and has also enriched the world of digital art and content creation. Here are some key takeaways from this project:

- Neural Style transfer builds on the fact to blend the content image to a style reference image such that the content is painted in the specific style
- Neural Style Transfer employs a pre-trained Convolutional Neural Network for feature extraction and separation of content and style representations from an image
- Neural Style Transfer network has two inputs: Content image and Style image. The content image is recreated as a newly generated image which is the only trainable variable in the neural network.
- The architecture of the model performs the training using two loss terms: Content Loss and Style Loss
- Image style transfer is a growing topic of research and aims to boost the artistic skills of every person. It is currently being used in diversified domains like gaming, virtual reality, photo and video editors etc

REFERENCES

REFERENCES

1. https://paperswithcode.com/paper/unpaired-image-to-image-translation-using -by Jun Yan Zhu,Taesung Park.
2. Leon A. Gatys, Alexander S. Ecker, Matthias Bethge.: A Neural Algorithm of Artistic Style. (2015).
3. https://github.com/alexjc/neural-enhance, Perpertual losses for Real-Time Style Transfer and Super- Resolution.

GitHub Link

1.